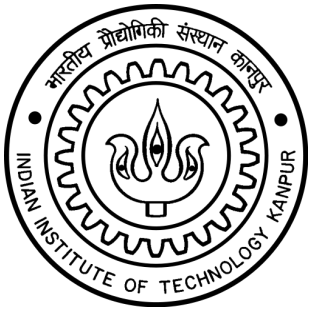


INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — RADIO ASTRONOMY LABORATORY

Experiment 2: RF Transmission, Reflections, and Standing Waves

---

Name: A.Aditya

Roll Number: 261310601

Programme: PhD

Department: SPASE

Date: August 19, 2026

1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

1.1 Objective

- Part A: To calculate the Phase velocity and velocity factor of the signal traveling in the coaxial cable frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable..
- Part B: Reflection coefficient and approximate characteristic impedance of the coaxial cable under test for open end and 50Ω load
- Part C: Inductance and capacitance per unit length of the cable using the phase velocity and the approx impedance of the cable (Z_0)
- Part D: To measure the insertion loss and attenuation per unit length of the coaxial cable.

2 Apparatus

- Digital storage oscilloscope (DSO)
- Radio Frequency(RF) signal generator
- Coaxial wire
- RF Tee connector
- 50 Ω load
- Spectral analyzer

3 Theory

In this experiment we will find the characteristic of coaxial cable which is used for transmission of electromagnetic wave where electric field and magnetic field are perpendicular to each other. A transmission line is a physical structure made for the transfer of signals from one place to another and coaxial cables are one of them. The structure of coaxial cable can be seen in Figure-1

When the signal is generated and travels through the cable, at the end of the cable it is left open and the open terminal provides theoretically infinite resistance to the the incoming wave which leads to the reflection of the wave. The reflected wave then interfere with the incoming wave constructively and destructively. As the digital storage oscilloscope (DSO) is connected to a single fixed point at the Tee junction, it cannot physically map the standing wave

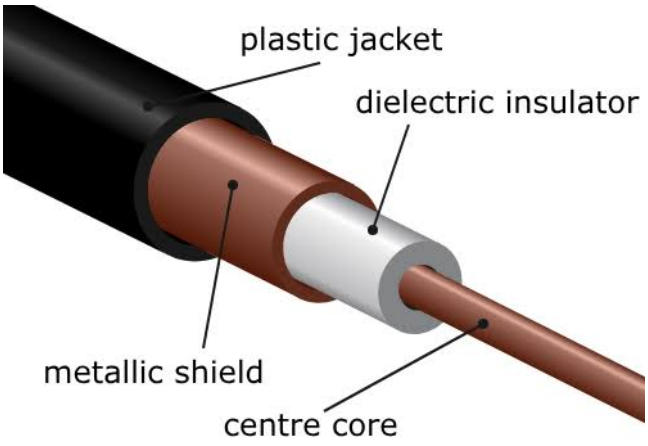


Figure 1: Coaxial wire structure

pattern along the length of the cable. However, as the signal generator's frequency is swept, the wave's electrical wavelength and its round-trip phase delay (ϕ_{RT}) change periodically. This causes the interference at the Tee junction to cycle between constructive phase (voltage maxima, $V_{\max}$) and destructive phase (voltage minima, $V_{\min}$), which are recorded by the oscilloscope.

Impedance is a factor which decides the smooth flow of the electrical signal in the medium. When the medium changes at the end of the wire with different impedance it might result in the wave to reflect back. The reflected wave will superimpose with the incoming wave and interfere constructively and destructively forming maxima and minima. The impedance also depends on the frequency.

3.1 Part A : Phase velocity and Velocity factor

Let the length of the coaxial cable be L, and let v be the phase velocity at which the electromagnetic wave propagates inside the cable. The phase velocity can be calculated using the relation:

$$v = 2L\Delta f \tag{1}$$

where Δf represents the frequency spacing between two consecutive voltage maxima or minima observed at the RF Tee connector.

After calculating the phase velocity v_p is calculated, it can be compared to the speed of light in a vacuum ($c \approx 3 \times 10^8$ m/s) to determine the dimensionless velocity factor (VF) of the cable:

$$\text{VF} = \frac{v_p}{c} \tag{2}$$

3.2 Part B Theory: Reflection Coefficient (Γ) and Characteristic Impedance (Z_0)

Reflection coefficient as the name suggest tells us about the amount of waves that bounce back at the end of the cable. The more the reflection the more the energy loss. It helps us determine the characteristic of the transmission line is suitable for transmission or not. The formula given by

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where Z_L is the load impedance terminating the end of the cable, and Z_0 is the characteristic impedance of the cable itself. This equation yields three critical scenarios

- Open circuit - The impedance Z_L of the load is infinite, then $\Gamma = 1$ which means 100% of the waves are reflected back.
- Short Circuit - The impedance Z_L of the load is 0, then $\Gamma = -1$ which means 100% of the waves are reflected back but with inverted phase.
- Matched load - The impedance Z_L of the load is equal to the cable impedance Z_0, then $\Gamma = 0$ which means all the energy is perfectly absorbed by the load, and there is zero reflection

At the Tee-zunction the oscilloscope reads the sum of the Reflected and incoming wave voltages -

$$V_{sum} = V_{reflected} + V_{incoming}$$

The magnitude of the reflection coefficient (Γ) is defined as the ratio of the reflected wave amplitude to the incident wave amplitude, we can estimate it directly from our oscilloscope's peak-to-trough ripple voltages -

$$\Gamma = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

By comparing the reflection coefficient of an open-circuited cable ($|\Gamma_{\text{open}}|$, which should theoretically be close to 1.0) with the reflection coefficient of a cable terminated with a 50Ω load ($|\Gamma_{50}|$:

If the termination matches the cable's characteristic impedance ($Z_0 \approx 50\ \Omega$), the reflected wave amplitude will drop significantly ($|\Gamma_{50}| \ll |\Gamma_{\text{open}}|$), proving the cable is matched and its internal impedance is close to 50 Ω.

3.3 Part C: Inductance(L) and Capacitance(C) per Unit Length

As we already know impedance of the material decided how fast of slow the signal can move in the transmission line, the coaxial cable consist of these parameters - Inductance(L) in Henries per meter (H/m) and capacitance (C) in Faraday per meter (F/m) which decides the smoothness and using the phase velocity and cable impedance we can calculate the Inductance and capacitance -

$$L = \frac{Z_0}{v_p} \tag{3}$$

$$C = \frac{1}{Z_0 v_p} \tag{4}$$

3.4 Part D: Cable Attenuation (α)

The loss of signal power during transmission in the real world is known as Attenuation. During transmission, two main kinds of losses occur in the cable:

- Skin effect: At higher frequencies, the current is pushed to the outer surface of the wire rather than penetrating deep into the conductor. This decreases the effective cross-sectional area, which increases resistance and leads to power loss.
- Dielectric losses: Due to the continuous alternating nature of the current, the surrounding dielectric material is continuously polarized in changing directions. This creates molecular friction and heat, causing additional power loss.

To measure this loss, we take the generated signal power and compare it with the power received in the spectrum analyzer which is connected to the generator via the coaxial cable. The total signal loss in decibels (dB) is called the insertion loss (IL)

$$\mathbf{IL} = P_{referance} - P_{cable}$$

4 Experiment setup

We will connect the signal generator and the oscilloscope to two ports of a Tee-connector, and attach the input of our coaxial cable under test to the third port,

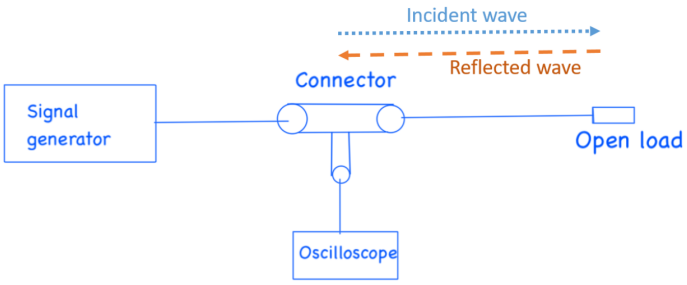


Figure 2: Experiment Setup

leaving its far end open. The frequency will be swept to find the spacing between consecutive maxima or minima on the oscilloscope, allowing us to calculate the phase velocity using the formula:

$$v_p = 2\ell\Delta f \tag{5}$$

Next, we will attach a 50 Ω load to the far end of the coaxial cable to determine the reflection coefficient ($|\Gamma_{50}|$) and verify if it matches the cable's characteristic impedance. Finally, we will measure the transmission power loss by connecting the signal generator to the spectrum analyzer, comparing the received power with and without the coaxial cable under test. Figure 2 represents the experiment setup.

5 Observation and Calculations

5.1 Part A: Standing Wave Voltage Sweep (Open-Ended Cable)

Coaxial cable length = 0.92 m.
Input frequency was swept from 57 MHz to 174 MHz while keeping the circuit end open. The resulting peak-to-peak voltage amplitudes measured at the Tee junction are recorded in Table 1.

Using the above observation we will calculate the frequency difference between two consecutive minima -

$$\Delta F = 174 - 57$$

$$\Delta F = 117 \text{ MHz}$$

Now using this we can calculate the Phase velocity -

Table 1: Voltage Amplitude vs. Frequency for Open-Ended Coaxial Cable

Frequency (MHz)	**Measured Voltage (V)**	**Frequency (MHz)**	**Measured Voltage (V)**
57	0.060 (Min)	115	1.580
62	0.225	120	1.540
68	0.578	125	1.500
70	0.693	130	1.420
75	0.974	135	1.330
80	1.280	140	1.230
85	1.510	145	1.080
90	1.610	150	0.932
95	1.710	155	0.762
97	1.720 (Max)	160	0.579
100	1.690	165	0.395
105	1.650	170	0.200
110	1.650	174	0.105 (Min)

The phase velocity (v_p) is calculated using the transmission line formula:

$$\begin{aligned} v_p &= 2\ell\Delta f \\ v_p &= 2 \times 92 \text{ cm} \times 117 \text{ MHz} \\ v_p &\approx 2.15 \times 10^8 \text{ m/s} \end{aligned}$$

Using the speed of light in a vacuum ($c \approx 3 \times 10^8$ m/s), the dimensionless **velocity factor** (VF) is determined by:

$$\begin{aligned} VF &= \frac{v_p}{c} \\ VF &= \frac{2.15 \times 10^8 \text{ m/s}}{3 \times 10^8 \text{ m/s}} \approx \mathbf{0.72} \end{aligned}$$

5.2 Part B: Reflection and Impedance Matching Observations

The maximum and minimum voltage ripple amplitudes were recorded over a complete standing-wave cycle for both the open-circuit configuration and when terminated with a 50 Ω load. These comparative observations are summarized in Table 2.

Table 2: Voltage Ripples for Open-Circuit vs. 50 Ω Terminated Configurations

Termination Type	$V_{\mathbf{max}}$ **(V)**	**Frequency at** $V_{\mathbf{max}}$	$V_{\mathbf{min}}$ **(V)**	**Frequency at** $V_{\mathbf{min}}$
Open Circuit	1.72	97 MHz	0.060	57 MHz
50 Ω Load	1.09	98 MHz	0.881	57 MHz

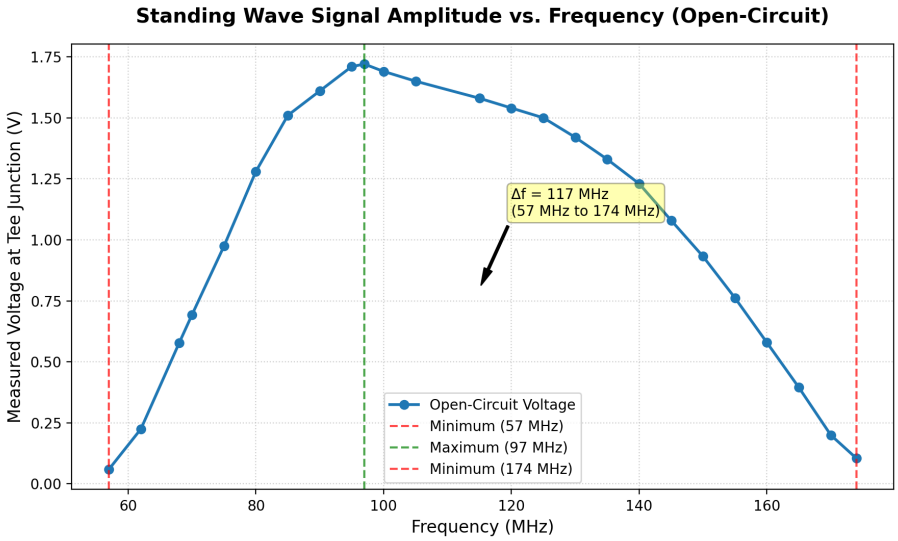


Figure 3: Measured standing-wave voltage amplitude as a function of frequency for the open-ended coaxial cable. The plot clearly shows the periodic wave variation between the consecutive minima at 57 MHz and 174 MHz.

5.2.1 Open-Circuit Reflection Coefficient

The maximum and minimum voltage amplitudes recorded during the open-circuit sweep are:

$$V_{\text{max,open}} = 1.72 \text{ V} \quad \text{(at 97 MHz)}$$
$$V_{\text{min,open}} = 60 \text{ mV} = 0.06 \text{ V} \quad \text{(at 57 MHz)}$$

The magnitude of the open-circuit reflection coefficient ($|\Gamma_{\text{open}}|$) is:

$$|\Gamma_{\text{open}}| = \frac{V_{\text{max,open}} - V_{\text{min,open}}}{V_{\text{max,open}} + V_{\text{min,open}}}$$
$$|\Gamma_{\text{open}}| = \frac{1.72 - 0.06}{1.72 + 0.06} = \frac{1.66}{1.78} \approx \mathbf{0.933}$$

5.2.2 Terminated 50 Ω Load Reflection Coefficient

With a matched 50 Ω termination connected to the far end of the coaxial line, the recorded voltage amplitudes are:

$$V_{\text{max,50}} = 1.09 \text{ V} \quad \text{(at 98 MHz)}$$
$$V_{\text{min,50}} = 881 \text{ mV} = 0.881 \text{ V} \quad \text{(at 57 MHz)}$$

The magnitude of the terminated reflection coefficient ($|\Gamma_{50}|$) is:

$$|\Gamma_{50}| = \frac{V_{\max,50} - V_{\min,50}}{V_{\max,50} + V_{\min,50}}$$
$$|\Gamma_{50}| = \frac{1.09 - 0.881}{1.09 + 0.881} = \frac{0.209}{1.971} \approx \mathbf{0.11}$$

5.2.3 Refined Characteristic Impedance (Z_0)

Using the terminated reflection coefficient, the actual characteristic impedance (Z_0) of the cable under test is calculated as:

$$Z_0 = 50 \left(\frac{1 - |\Gamma_{50}|}{1 + |\Gamma_{50}|} \right)$$
$$Z_0 = 50 \left(\frac{1 - 0.11}{1 + 0.11} \right)$$
$$Z_0 = 50 \left(\frac{0.89}{1.11} \right) \approx \mathbf{40.1\ \Omega}$$

5.3 Part C: Inductance and Capacitance per Unit Length Calculations

Using the measured phase velocity ($v_p = 2.15 \times 10^8$ m/s) and the refined characteristic impedance ($Z_0 = 40.1\ \Omega$), the distributed line parameters are calculated:

5.3.1 Inductance per Unit Length (L')

$$L' = \frac{Z_0}{v_p}$$
$$L' = \frac{40.1\ \Omega}{2.15 \times 10^8\ \text{m/s}}$$
$$L' \approx \mathbf{1.865 \times 10^{-7}}\ \text{H/m} = \mathbf{186.5}\ \text{nH/m}$$

5.3.2 Capacitance per Unit Length (C')

$$C' = \frac{1}{Z_0 v_p}$$
$$C' = \frac{1}{40.1\ \Omega \times (2.15 \times 10^8\ \text{m/s})}$$
$$C' = \frac{1}{8.6215 \times 10^9}$$
$$C' \approx \mathbf{1.160 \times 10^{-10}}\ \text{F/m} = \mathbf{116.0}\ \text{pF/m}$$

5.4 Part D: Coaxial Cable Attenuation Observations

Power loss measurements were performed using a spectrum analyser. With a stable reference power baseline established at $P_{\text{ref}} = 20.0$ dBm using a short reference cable, the transmitted power (P_{cable}) was measured across four key high-frequency test points. The raw experimental readings are presented in Table 3.

Table 3: Transmitted Power Level Observations for Attenuation Analysis

Test Frequency (GHz)	**Reference Power P_{ref} (dBm)**	**Received Power P_{cable} (dBm)**
0.5	20.0	18.50
1.0	20.0	17.45
1.6	20.0	17.00
2.0	20.0	16.49

Insertion loss (IL) and attenuation per unit length (α) are evaluated across four high-frequency test points with a reference power baseline of $P_{\text{ref}} = 20$ dBm and a cable length of $\ell = 0.92$ m ($\ell = 92$ cm):

1. **At** 0.5 **GHz** ($P_{\text{cable}} = 18.50$ **dBm):**

$$IL = 20 - 18.50 = 1.50 \text{ dB}$$
$$\alpha_{\text{cm}} = \frac{1.50 \text{ dB}}{92 \text{ cm}} \approx \mathbf{0.016} \text{ dB/cm}$$
$$\alpha_{\text{m}} = \frac{1.50 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{1.63} \text{ dB/m}$$

2. **At** 1.0 **GHz** ($P_{\text{cable}} = 17.45$ **dBm):**

$$IL = 20 - 17.45 = 2.55 \text{ dB}$$
$$\alpha_{\text{cm}} = \frac{2.55 \text{ dB}}{92 \text{ cm}} \approx \mathbf{0.028} \text{ dB/cm}$$
$$\alpha_{\text{m}} = \frac{2.55 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{2.77} \text{ dB/m}$$

3. **At** 1.6 **GHz** ($P_{\text{cable}} = 17.00$ **dBm):**

$$IL = 20 - 17.00 = 3.00 \text{ dB}$$
$$\alpha_{\text{cm}} = \frac{3.00 \text{ dB}}{92 \text{ cm}} \approx \mathbf{0.033} \text{ dB/cm}$$
$$\alpha_{\text{m}} = \frac{3.00 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{3.26} \text{ dB/m}$$

4. **At 2.0 GHz ($P_{\text{cable}} = 16.49$ dBm):**

$$IL = 20 - 16.49 = 3.51 \text{ dB}$$
$$\alpha_{\text{cm}} = \frac{3.51 \text{ dB}}{92 \text{ cm}} \approx \mathbf{0.038} \text{ dB/cm}$$
$$\alpha_{\text{m}} = \frac{3.51 \text{ dB}}{0.92 \text{ m}} \approx \mathbf{3.82} \text{ dB/m}$$

6 Conclusion

In this laboratory experiment, the wave propagation, reflection, and attenuation characteristics of a 0.92 m coaxial transmission line were successfully investigated. The key parameters determined from our experimental measurements are summarized as follows:

- **Phase Velocity and Velocity Factor:** The frequency spacing between consecutive voltage minima ($\Delta f = 117$ MHz) yielded a signal phase velocity of $v_p = 2.15 \times 10^8$ m/s. This corresponds to a dimensionless velocity factor (VF) of **0.72**.
- **Reflection Coefficient and Matching Verification:** For the open-circuit termination, the reflection coefficient magnitude was $|\Gamma_{\text{open}}| \approx 0.933$. When terminated with a 50 Ω load, this dropped significantly to $|\Gamma_{50}| \approx 0.11$. This drastic reduction verifies that the cable is approximately matched to a 50 Ω system, confirming that the cable's characteristic impedance (Z_0) is close to 50 Ω (with our refined calculation yielding 40.1 Ω).
- **Line Parameters per Unit Length:** Using the refined characteristic impedance ($Z_0 = 40.1$ Ω), the distributed inductance and capacitance per unit length of the cable were determined to be $L' = 186.5$ nH/m and $C' = 116$ pF/m, respectively.
- **Cable Attenuation:** The attenuation per unit length (α) was shown to be highly frequency-dependent, increasing steadily from 1.63 dB/m at 0.5 GHz to 3.82 dB/m at 2.0 GHz.

6.1 Sources of Measurement Uncertainty

To maintain scientific rigor, several potential sources of experimental uncertainty must be considered:

1. **Instrument Resolution and Wave Flatness:** Near the exact voltage peaks (V_{max}) and troughs (V_{min}), the sinusoidal voltage curve becomes extremely flat. Identifying the precise frequency of the turning points introduces minor visual reading errors on the oscilloscope.

2. **Connector and Adapter Losses:** The Tee connector and cable junctions introduce small parasitic capacitance, inductances, and impedance discontinuities. These non-ideal junctions cause minor reflections and signal dissipation that are not accounted for in ideal transmission line equations.

3. **Physical Length Measurements:** Measuring the cable length ($\ell = 0.92$ m) with a standard inch-tape introduces a small physical tolerance (± 0.5 cm), which directly propagates into the phase velocity and attenuation calculations.

4. **Impedance Mismatches in the Source and Load:** Slight deviations of the generator's source impedance or the termination cap from the nominal 50 Ω value prevent absolute impedance matching, explaining why $|\Gamma_{50}|$ did not reach exactly zero.

7 AI Usage Declaration

The following uses of Artificial Intelligence tools (specifically, LLM assistance) were made during the preparation of this report:

1. **LaTeX Formatting and Document Structure:** AI was used as a guide to understand LaTeX formatting, helping to structure the document, set up packages, manage margins, and debug compilation errors to ensure a clean and professional presentation.

2. **Polishing and Refining Written Content:** AI was used in correcting the grammar.

3. **Conceptual Clarification:** AI was used as a supplementary tool to clarify theoretical concepts discussed during the laboratory, such as how oscilloscope detects the wave or impedance relation with phase velocity.